Data Analysis Intern Interview Questions

Excel, SQL, Python, Pandas, EDA & Power BI

Interview Structure

This version is focused on practical questions that commonly appear in data analyst interviews, including questions reported in current interview-preparation sources and real interview-experience discussions. It is intentionally written as interview questions rather than textbook definitions. ■cite■turn0search0■turn0search4■turn0search9■

1. Excel — Questions Actually Used in Analyst Interviews

1. What is the difference between VLOOKUP, XLOOKUP, and INDEX-MATCH? When would you choose each one?
2. You have two Excel sheets: one contains Customer ID and Name, and the other contains Customer ID and Sales. How would you bring the sales value into the first sheet?
3. How would you find duplicate customer records in Excel?
4. What is the difference between COUNT, COUNTA, COUNTIF, and COUNTIFS?
5. How would you calculate total sales for a particular region and month using SUMIFS?
6. What is a PivotTable, and how would you use one to find the top 5 products by revenue?
7. You have 500,000 rows of data and the workbook is very slow. What would you do?
8. How would you identify missing values and inconsistent entries in an Excel dataset?
9. Explain relative, absolute, and mixed cell references with an example.
10. How would you create a month-over-month sales growth calculation in Excel?
11. You have sales data with dates stored as text. How would you convert them into proper dates?
12. How would you use conditional formatting to highlight sales below the monthly target?
13. Can you explain a situation where you used Excel to solve a real business problem?

2. SQL — Query-Based Interview Questions

1. Write a query to find the second-highest salary from an Employee table.
2. Write a query to find the top 3 highest-paid employees in each department.
3. What is the difference between WHERE and HAVING? Give an example where both are used.
4. Explain INNER JOIN and LEFT JOIN. When would a LEFT JOIN return more rows than an INNER JOIN?
5. Write a query to find customers who have never placed an order.
6. Write a query to find duplicate records in a table.
7. Write a query to calculate total sales for each product category.
8. Write a query to calculate monthly revenue from an Orders table.
9. What is the difference between ROW_NUMBER(), RANK(), and DENSE_RANK()?
10. Write a query to find employees whose salary is greater than the average salary of their department.
11. What is a CTE? Rewrite a subquery using a CTE.
12. What is the difference between UNION and UNION ALL?
13. How would you find the first and most recent order placed by each customer?
14. Write a query using LAG() to compare a customer's current purchase with their previous purchase.

15. A query is taking several minutes to run. How would you investigate and improve its performance?

16. Explain the order in which SQL clauses such as FROM, WHERE, GROUP BY, HAVING, SELECT, and ORDER BY are logically processed.

3. Python — Data Analyst Interview Questions

1. What Python data structures do you commonly use while working with datasets?
2. What is the difference between a list, tuple, set, and dictionary?
3. How would you remove duplicates from a list while preserving the original order?
4. Write a Python function to count the frequency of each word in a sentence.
5. How would you handle missing values in a Python dataset?
6. What is the difference between == and is?
7. What is a list comprehension? Rewrite a simple loop using list comprehension.
8. How would you read a CSV file and inspect its columns using Python?
9. What is the difference between a Python list and a NumPy array?
10. Why is vectorization preferred over repeatedly looping through large datasets?
11. Write a function that takes a list of numbers and returns the mean, minimum, and maximum.
12. You receive a column containing values such as ' Delhi ', 'delhi', 'DELHI'. How would you standardize it?
13. How would you handle an error while converting a string column into numeric values?
14. Explain a Python data-analysis project you have worked on and the steps you followed.

4. Pandas — Practical Interview Questions

1. How do you read a CSV file into a Pandas DataFrame?
2. What is the difference between loc and iloc?
3. How would you select all rows where Sales is greater than 10,000?
4. How would you find the number of missing values in every column?
5. How would you remove rows containing missing values?
6. When would you use fillna() instead of dropna()?
7. How would you remove duplicate rows from a DataFrame?
8. How would you convert a column to datetime?
9. How does groupby() work? Write a solution to calculate total sales by category.
10. How would you find the top 5 customers by total spending?
11. What is the difference between merge(), join(), and concat()?
12. How would you merge customer and order DataFrames using Customer_ID?
13. How would you create a new Profit column from Revenue and Cost?
14. How would you sort a DataFrame by Sales in descending order?
15. How would you find the average order value for each city?
16. You are given a messy dataset. Walk through the Pandas steps you would perform before starting EDA.

5. EDA — Questions Based on Real Analysis Tasks

1. You are given a completely new dataset. What is the first thing you would do?
2. Which checks do you perform to understand the structure and quality of a dataset?
3. How would you identify missing values, duplicates, incorrect data types, and inconsistent categories?
4. When would you use mean, median, or mode to describe a variable?
5. How would you detect outliers using the IQR method?
6. How would you decide whether an outlier should be removed or retained?
7. What is the difference between correlation and causation?
8. How would you investigate whether two numerical variables are related?
9. Which plots would you use to analyze a numerical variable and why?
10. How would you analyze the distribution of a categorical variable?
11. You notice that sales suddenly dropped in one month. How would you investigate the reason?
12. How would you identify the most important factors associated with customer churn?
13. How would you check whether a dataset contains duplicate customers?
14. After completing EDA, how would you convert your findings into business recommendations?
15. Explain one EDA project from raw data to final insights.

6. Power BI — Interview Questions

1. What is the difference between Power BI Desktop and Power BI Service?
2. What is Power Query, and what type of work would you perform in it?
3. What is the difference between Power Query and DAX?
4. What is the difference between a calculated column and a measure?
5. Why are measures generally preferred for aggregations?
6. What is filter context in Power BI?
7. What is the difference between a slicer and a filter?
8. How do relationships work between tables in Power BI?
9. What is a star schema, and why is it recommended for Power BI models?
10. What is the difference between a fact table and a dimension table?
11. Write a basic DAX measure to calculate Total Sales.
12. What does CALCULATE() do in DAX?
13. What is the difference between SUM() and SUMX()?
14. How would you calculate year-over-year sales growth?
15. You have a dashboard with 10 visuals and it is loading slowly. How would you troubleshoot it?
16. How would you design a sales dashboard for a business manager? Which KPIs would you show?
17. A stakeholder says the dashboard numbers do not match Excel. How would you investigate the discrepancy?
18. How would you make a Power BI report easier for a non-technical user to understand?

7. Data Cleaning & Business Case Questions

1. You receive a sales file where 8% of customer ages are missing, dates have multiple formats, and region names are inconsistent. How would you clean it?

2. A manager says revenue is down 15% this month. What questions would you ask before explaining why?

3. Sales increased by 20%, but profit decreased by 5%. How would you investigate this?

4. One product has very high sales but almost no profit. What would you check?

5. A dashboard shows a sudden spike in orders. How would you determine whether it is a real business change or a data-quality issue?

6. How would you validate your analysis before presenting it to management?

7. How would you explain a technical data finding to someone who does not know SQL or Python?

8. If two data sources give different values for the same KPI, how would you determine which one is correct?

8. Practical Interview Tasks

1. Given an Excel sales dataset, create a PivotTable showing revenue by region and month.

2. Given two Excel sheets, use a lookup to combine customer information with transaction data.

3. Given an SQL Orders table, write queries for total revenue, monthly revenue, top customers, and repeat customers.

4. Given a Pandas DataFrame, clean missing values, remove duplicates, group by category, and identify the highest-revenue category.

5. Given a messy dataset, perform EDA and present 3–5 meaningful findings.

6. Given a cleaned dataset, build a Power BI report with KPIs, trends, category analysis, and filters.

7. Given a business problem such as declining sales, use SQL/Python/Power BI to investigate and explain your conclusion.

9. Suggested Interview Evaluation

1. Excel: Can the candidate solve lookup, aggregation, cleaning, PivotTable, and reporting problems without relying only on memorized formulas?

2. SQL: Can the candidate write correct queries, reason about joins and aggregations, and solve practical business questions?

3. Python/Pandas: Can the candidate manipulate real data, clean it, aggregate it, and explain the code?

4. EDA: Can the candidate identify data-quality issues, patterns, anomalies, relationships, and useful insights?

5. Power BI: Can the candidate build a sensible model, write basic DAX, use filters correctly, and communicate insights?

6. Business Thinking: Can the candidate move from 'what happened?' to 'why did it happen?' and 'what should we do next?'

7. Communication: Can the candidate clearly explain assumptions, methodology, limitations, and conclusions?